DO NOW — RECALL

Fetch–decode–execute and buses

Name: _____ Class: _____ Date: _____

1. Which register holds the address accessed during a fetch?

2. What does the CIR hold?

3. What kind of information travels along the address bus?

4. A rising-edge D flip-flop has Q=0. D changes to 1 between clock edges; the next rising edge occurs while D=1. State Q before and after that edge, and explain why.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

MAR.

2. Model answer

The current instruction that is being decoded or executed.

3. Model answer

Addresses identifying the memory or device location.

4. Model answer

Before edge Q stays 0; at rising edge D=1 is sampled so Q becomes 1. It changes on the active edge, not merely when D changes.

Vocabulary — know these terms

fetch — Retrieve next instruction from memory.

decode — Interpret opcode and operands.

execute — Perform the operation.

address bus — Transfers memory/device addresses.

data bus — Transfers data and instructions.

control bus — Transfers signals coordinating reads and writes.

NEW LEARNING

Read, annotate and explain

Fetch uses the buses

The address bus carries the address selected by MAR; memory returns instruction data on the data bus; control signals coordinate the transfer.

Decode interprets instruction

The control unit examines CIR to decide which operation and operands are involved.

Execute changes state

An instruction may update ACC or memory, or alter PC for a jump. Interrupt handling is coordinated at appropriate points and can temporarily redirect execution.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Fetch the instruction at address 40 and explain what each bus contributes.

Predict first

MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

Explain the answer

The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For each bus—address, data and control—state its main function and give one example of the information transported.

Question 2

Trace a fetch starting PC=64, then explain what happens when CIR contains JMP 100.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare what happens in the instruction cycle during a normal arithmetic instruction and while responding to an interrupt.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An unconditional branch can change the Program Counter away from its sequential value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.